# Fingerprints, then deeper

Fingerprints Up Close

Name _____ Date _____



Little Wonderer held up a glass. "I don't see anything," the little one said. Big Wonderer tipped it to the light. A faint print showed up. Your fingertips have tiny ridges. The ridges carry a thin film of sweat and oil. When you touch something, the film copies the ridge pattern. Most prints are latent, which means hidden until someone makes them show.



Most prints belong to one of three pattern families. In a loop, ridges come in and curve back out the same side. In a whorl, ridges make circles or a spiral. In an arch, ridges flow across with a bump in the middle. Loops are the most common. Arches are the rarest. No two prints have ever been found to match, not even on identical twins.



Prints form before a baby is born. At about 7 to 8 weeks of growing, puffy pads appear on the fingertips. At about 10 to 11 weeks, ridges start to form. By about 15 to 17 weeks, the pattern is set. Twins get different prints because each fingertip grows in its own tiny way. After that, the pattern normally just grows bigger and keeps its shape, though a deep cut or some skin diseases can leave a scar.

| | |
|---|---|
| **ridge** | a tiny raised line of skin on your fingertip |
| **latent** | hidden until someone makes it show |
| **whorl** | a print pattern with circles or a spiral |
| **pore** | a tiny opening in the skin where sweat comes out |
| **bifurcation** | a place where one ridge splits in two |
| **minutiae** | tiny places where a ridge ends or splits |

# Fingerprints, then deeper

Fingerprints Up Close

Name _____ Date _____



Koalas have prints that look very much like ours. Koalas are only distantly related to us, and their cousins, wombats and kangaroos, have none. Scientists think prints formed separately in both because both grip things. What are ridges for? In one lab test, ridges made the shakes from rubbing a fine texture much stronger. So scientists think ridges help us feel fine textures.



Detectives make hidden prints show up in a lab. They dust powder, use a chemical called ninhydrin that turns prints purple, or use superglue fumes. Those are jobs for trained lab workers, not for kids. Little Wonderer and Big Wonderer made a safe one. They rubbed a pencil on paper, pressed a fingertip on it, and lifted the print with clear tape. The big one found a spot where a ridge split in two.



After a long bath, fingers get wrinkly. Your nerves cause it. Blood vessels under the skin get narrower, and the skin puckers. Fingers with cut nerves do not wrinkle. Maybe the wrinkles help us grip wet things, like treads on a tire. One study found people moved wet marbles faster with wrinkly fingers. Another found no difference. So for now, we say maybe.

1. Why can a fingertip leave a hidden print on a glass?
2. How do fingerprints form, and why do identical twins still have different prints?
3. What might ridges and wrinkly fingers help us do, and why do scientists say maybe?

## Your turn

Make a pencil-and-tape print. Write one sentence about its pattern.

Example: My print is a loop because the ridges curve back out the same side.